UE26MA141A Engineering Mathematics – I · Unit 2 · Higher-Order Linear Differential Equations

Graphs and diagrams

Every graph and diagram in scope, redrawn as clean vector figures.

Class 004 · Particular integral Type 2 (sin ax, cos ax) · compiled 16 October 2026

ESA ★★★	Asked repeatedly in past ESA papers
ESA ★★	Asked in past ESA papers or computed in class-work problems
ESA ★	Conceptual: taught in class, not examined so far
ADDED FOR CLARITY	Standard context that is not in the class material
TRANSCRIPT ONLY	Said in the lecture but not on any slide
VIDEO ONLY	Seen only in a frame of the class video
NOT ON CURRENT SLIDES	From older material or past papers, not this year's slides
S004·9	Unit 2 deck 004 (“Particular integral of Type 2”), slide 9 (rendered .pptx)
T004 00:10:32	Unit 2 AV-summary transcript “Particular integral of standard functions – Type 2” (24.7 min), segment start
T004b 00:08:13	Unit 2 AV-summary transcript “Particular integral Type 1 and Type 2” (12.6 min, newer recording); its Type 1 part is tagged T004a in the class 003 pack
CW U2 P13	Unit 2 class-work problem 13
ESA Dec-23 3a	Past paper: UE22MA141A Dec 2023, question 3.a
ESA Jan-24 s1 2d	Past paper: UE23MA141A Jan 2024 set 1, question 2.d

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3. Figure G2.2 · The failure case grows: $x\cos 2x$ against a bounded PI
4. Figure G3.1 · $\sin^2 x$ is a constant plus a cosine of double frequency

The deck has no graphs, so every figure is added for clarity. Colours follow the Okabe–Ito order: blue = the right-hand side or the no-failure case, vermilion = the particular integral or the failure case, green = a constant part; grey dashed lines are references (envelopes, the value a at which $f(-a^2)=0$). Every plotted PI was checked by substituting it back with sympy.

1 The Type 2 rule: replace D^2 by $-a^2$

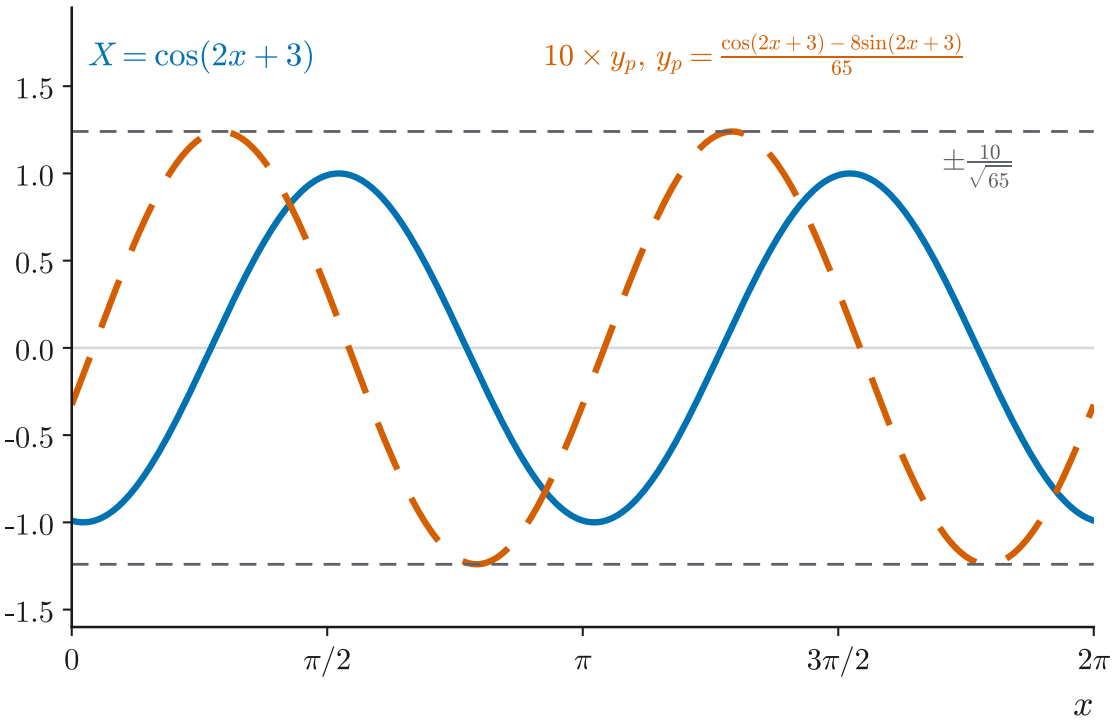


Figure G1.1. Same frequency, new amplitude and phase: the PI of $(D^3+1)y = \cos(2x+3)$. ESA ★★

ADDED FOR CLARITY S004·8 · S004·9

How to read it. The forcing $\cos(2x+3)$ (blue) and ten times its PI $\frac{\cos(2x+3) - 8\sin(2x+3)}{65}$ (vermilion dashed). Both have period π; the PI is a single sinusoid of amplitude $\frac{\sqrt{1+64}}{65} = \frac{1}{\sqrt{65}}$ (grey lines, scaled by 10) shifted in phase. This is what the rationalised answer $\cos - 8\sin$ means.

2 Case of failure: $f(-a^2) = 0$

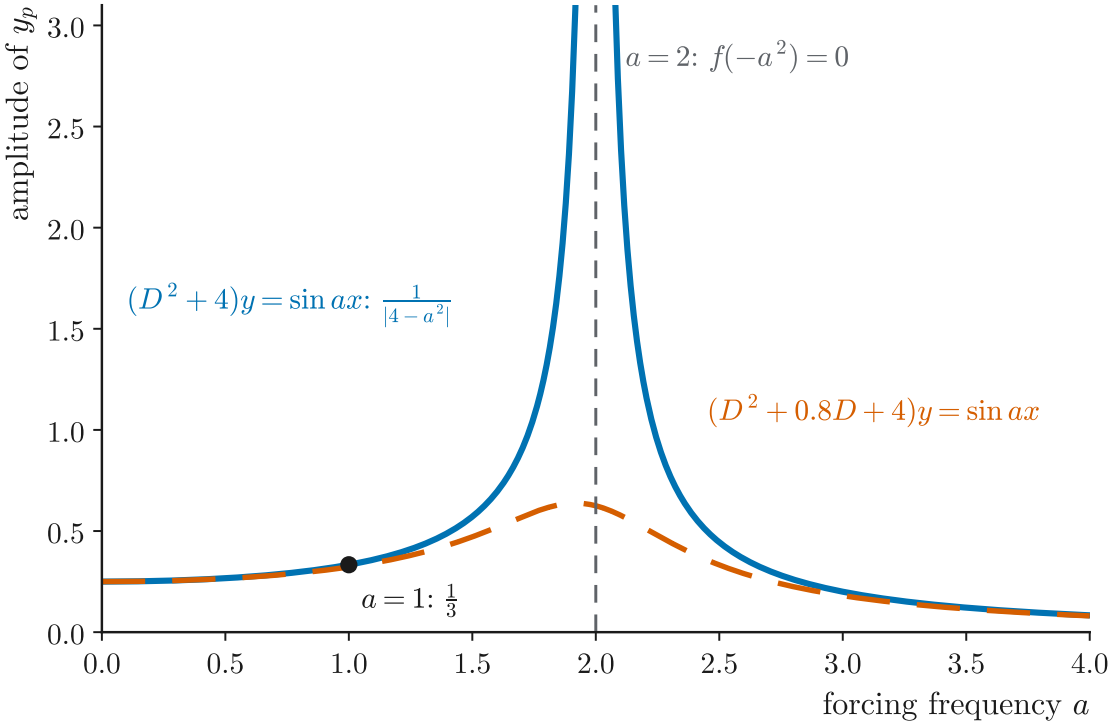


Figure G2.1. Amplitude of the PI, $\frac{1}{|f(-a^2)|}$, against the forcing frequency a. ESA ★★★

ADDED FOR CLARITY S004·4 · S004·5

How to read it. For $(D^2 + 4)y = \sin ax$ the PI is $\frac{\sin ax}{4 - a^2}$, of amplitude $\frac{1}{|4 - a^2|}$ (blue). It is $\frac{1}{4}$ at $a = 0$, $\frac{1}{3}$ at $a = 1$ (black dot) and blows up at $a = 2$ (grey line), where $f(-a^2) = 0$ and the Type 2 rule fails. With a damping term $(D^2 + 0.8D + 4$, vermilion dashed) $f(-a^2) = 4 - a^2 + 0.8D$ never vanishes and the peak stays finite (resonance).

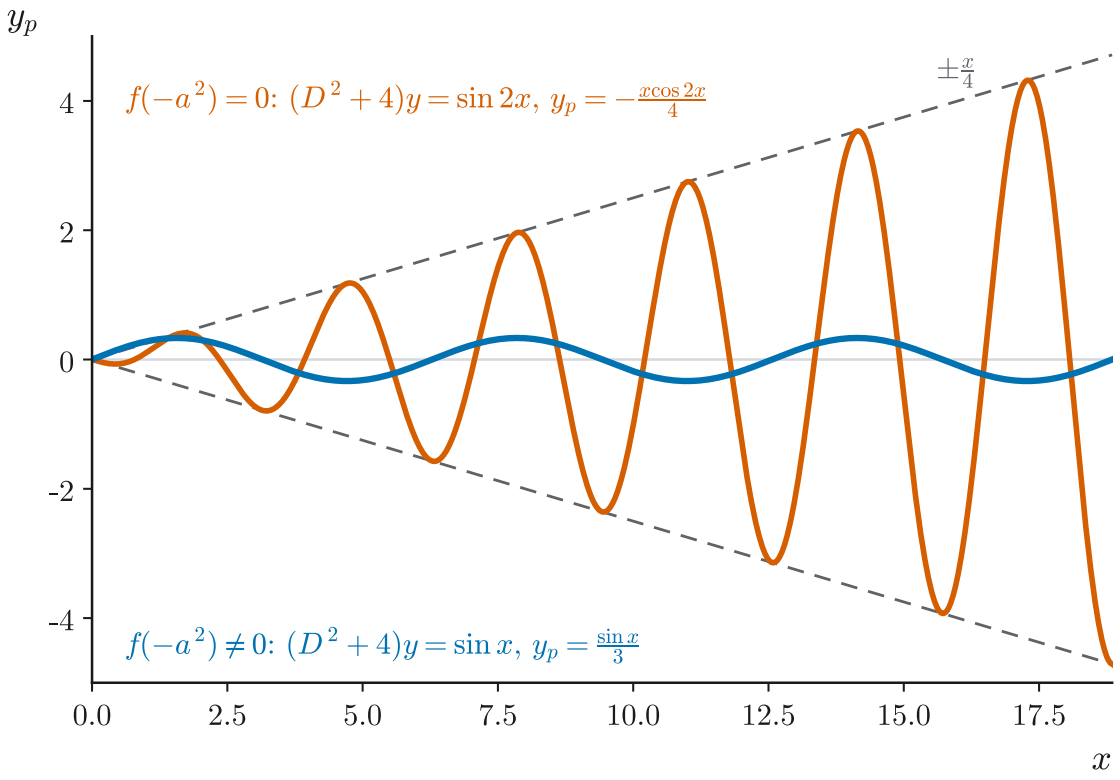


Figure G2.2. The failure case grows: $x\cos 2x$ against a bounded PI. ESA ★★★

ADDED FOR CLARITY S004·5 · S004·16

How to read it. With $(D^2+4)y=\sin 2x$ the frequency 2 is in the CF, so the PI is $-\frac{x\cos 2x}{4}$ (vermilion): an oscillation whose envelope $\pm\frac{x}{4}$ (grey dashed) grows linearly. With $\sin x$ instead ($f(-1)=3\neq 0$) the PI $\frac{\sin x}{3}$ (blue) stays bounded.

3 Products, powers and sums: reducing to Type 2

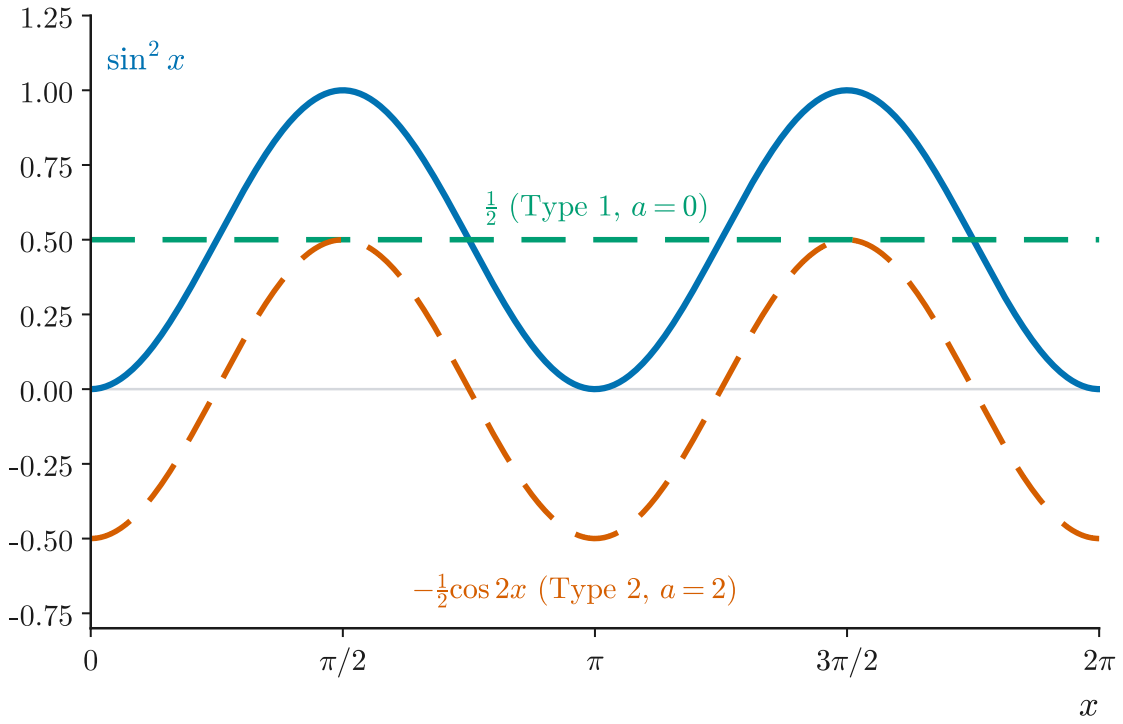


Figure G3.1. $\sin^2 x$ is a constant plus a cosine of double frequency. ESA ★★

ADDED FOR CLARITY S004·7 · S004·17

How to read it. $\sin^2 x$ (blue) never goes negative and has period π. It is the constant $\frac{1}{2}$ (green dashed, a Type 1 term with $a = 0$) plus $-\frac{1}{2}\cos 2x$ (vermilion dashed, a Type 2 term with $a = 2$, not 1). Each part gets its own rule; the two PIs are added.